

Jacob Lee

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Education

California Polytechnic State University

B.S. Computer Science, Concentration in ML/AI; Minor in Linguistics GPA: 3.9

San Luis Obispo

June, 2027

Coursework: Software Engineering, Design and Analysis of Algorithms, Programming Languages, Systems Programming, Computer Security, Databases

Languages: Python, TypeScript, Java, C, SQL

Frameworks/Libraries: React, React Native, Tailwind CSS, Pandas, pytest

Databases & Backend: MongoDB, PostgreSQL, Supabase

Developer Tools: Git, GitHub

Experience

CodeBox Mustang Maps Project

Software Engineer

San Luis Obispo, CA

October 2025 – June 2026

- Developed responsive UI features and campus-location classifications for Mustang Maps, a Cal Poly-focused mapping application for classes, events, facilities, and utilities, using React, TypeScript, Tailwind CSS, and HTML/CSS.
- Wrote pytest tests for multi-classification search and filtering logic, validating that combined filters returned only matching campus locations and handled empty-result queries correctly.
- Contributed to the frontend/design subteam by translating interface designs into reusable components and improving consistency across filter, listing, and search screens.
- Collaborated with a large software engineering team using Git and GitHub for version control and shared development workflows.

TOMS Accessibility Project with 40 Prado Homeless Services Center (CAPSLO)

Software and Hardware Developer

San Luis Obispo, CA

March 2025 – June 2026

- Programmed and integrated Arduino microcontrollers to develop an automatic door system improving accessibility of the facility and services for individuals with disabilities.
- Modeled mechanical components and linkage mechanisms, wrote control software, and conducted on-site assessments to ensure accurate installation and reliable performance.

Algoverse

Reinforcement Learning Researcher

San Francisco, CA

June 2026 – Present

- Built a Gymnasium-compatible POMDP benchmark with 2 latent states, 2 actions, and four controlled failure conditions to separate reward misspecification from observation insufficiency.
- Developed reproducible Q-learning and PPO pipelines across 4 conditions × 5 paired seeds, with passive and privileged telemetry for policy, value, TD-error, reward, and trajectory analysis.

Cal Poly

Volunteer Undergraduate Research Assistant - Cal Poly Agentic AI Team

San Luis Obispo, CA

August 2025 – February 2026

- Contributed to an AI-driven multi-agent search-and-rescue system for target prioritization and route planning.
- Evaluated deployment across AWS, Azure, and GCP for performance, scalability, and cost tradeoffs.

Kaiser Permanente

Pharmacy Administration Assistant Intern

Sacramento, CA

June 2022 – May 2024

- Automated onboarding workflows for cybersecurity protocols and supported data-security procedures across KP Sacramento Valley, **reducing manual setup time by 40%**.

Interests: Board games, magic, pickleball, learning Spanish, baking